scGPT Human Validation — Reviewer 2 (Blinded)

Purpose

Independently determine how each citing paper uses scGPT. Machine labels, confidence, agent outputs, and machine-selected evidence are intentionally hidden.

Workflow

Calibration: complete CAL-01 to CAL-03 independently, then discuss those three cases and freeze the codebook.
Formal validation: complete every HV row in the displayed order. Do not open the other reviewer document or the consensus document until both formal reviews are locked.
Reconciliation: after both reviewers lock their answers, the coordinator copies both records into the consensus document. Discuss only disagreements. Use REFERENCE_UNRESOLVED when agreement is not possible.
Evidence boundary: use only the linked approved article versions. Search all occurrences of “scGPT” and inspect the surrounding methods, results, figures, tables, and supplements that are linked.

Allowed values

Citation validity: Yes / No / Unclear
Method executed: Yes / No / Unclear
Primary use: Mention only / Benchmark / Extend or develop / Apply to biological data / Other / Unclear
Biological insight: Yes / No / Unclear
Evidence sufficiency: Sufficient / Incomplete / Citation not located
Supporting evidence: exact quote plus section, page, paragraph, figure, or table locator.

Review table

Article and approved evidence links	Reviewer 2 independent record
CAL-01 | CALIBRATION **scMGCL: accurate and efficient integration representation of single-cell multi-omics data** 2025 | Bioinformatics DOI: 10.1093/bioinformatics/btaf392 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: In our study, we applied scGPT to generate embeddings for the PBMCMultiome dataset and concatenated them with the original features as input to scMGCL for multi-omics integration (Fig. S22, available as supplementary data at Bioinformatics online). Compared to the original unintegrated data (Before Integration; Fig. S4, available as supplementary data at Bioinformatics online), the scGPT-derived embeddings more effectively represent joint characteristics of the RNA and ATAC modalities (Fig. S22A, available as supplementary data at Bioinformatics online), as reflected by consistent topological structures and aligned cell-type distributions. Notes:
CAL-02 | CALIBRATION **Genome language modeling (GLM): a beginner's cheat sheet** 2025 | Biology Methods and Protocols DOI: 10.1093/biomethods/bpaf022 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: whereas, scGPT [69] models single cell RNA expression profiles Notes:

CAL-03 | CALIBRATION **Building a learnable universal coordinate system for single-cell atlas with a joint-VAE model** 2024 | Communications Biology DOI: 10.1038/s42003-024-06564-0 Article page Full text 1 Full text 2	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Furthermore, as an interpretable model, UniCoord is complemented with the fashionable large-scale pretrained models Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-022 | FORMAL VALIDATION **PantheonOS: An Evolvable Multi-Agent Framework for Automatic Genomics Discovery** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.26.707870 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop / Apply to biological data Biological insight: Yes Evidence sufficiency: Sufficient Supporting quote + locator: To illustrate the router's utility in a biologically meaningful application, we used PantheonOS to perform in-silico gene perturbation analysis on the human fetal heart single-cell data using scGPT43, which was automatically selected by the router for this perturbation task. Perturbation effect sizes for 241 heart disease-associated genes38,40 reveal that cardiac structural genes, MYH6, MYH7, MYL2, DES, and ACTC1, exhibit the largest embedding shifts upon both knockout (KO) and overexpression (OE), consistent with their central roles in cardiomyocyte identity and function Notes:
HV-028 | FORMAL VALIDATION **Single-cell sequencing in precision management of chronic myeloid leukemia** 2026 | Journal of Translational Genetics and Genomics DOI: 10.20517/jtgg.2026.42 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Foundation models such as scGPT further illustrate how large-scale pretrained representations may support cell-type annotation, perturbation modeling, and cross-dataset transfer learning, although their reliability inLi *et al*. *J Transl Genet Genom.* 2026;10:386-400 Page 396 rare hematologic malignant states still requires benchmarking and interpretability assessment[65]. Notes:

HV-012 | FORMAL VALIDATION **MOSAIC: Intra-tumoral heterogeneity characterization through large-scale spatial and cell-resolved multi-omics profiling** 2025 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.05.15.654189 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: The existing literature on analysis methods for snRNAseq is already rich and includes methods for processing the data (see, e.g., the literature review in Heumos et al. 2023), differential expression analysis (see Squair et al. and references therein), clustering methods (see Kiselev et al. 2019 and references therein), and recently foundation models pre-trained on large scale data sets (Yang et al. 2022, Zhang et al. 2022, Hao et al. 2024, Cui et al. 2024). Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-048 | FORMAL VALIDATION **scBaseCount: an AI agent-curated, uniformly processed, and autonomously updated single cell data repository** 2025 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.02.27.640494 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: d disease mechanisms, while also providing valuable training data for AI-driven modeling of cellular states. These models are designed to capture context-dependent cellular function and behavior and predict cell state in experimentally unobserved settings such as in response to perturbation (Lopez et al., 2018; Theodoris et al., 2023; Cui et al., 2024; Hao et al., 2024; Rosen et al., 2023; Pearce et al., 2025; Adduri et al., 2025). Notes:
HV-005 | FORMAL VALIDATION **A multimodal cell-free RNA language model for liquid biopsy applications** 2025 | Nature Machine Intelligence DOI: 10.1038/s42256-025-01148-x Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Single-cell foundation models have also made prominent strides in transcriptomics. Models such as scBERT19, Exceiver20, GeneCompass21, scFoundation22, Geneformer23, scGPT24 and UCE25 have demonstrated efficacy in tasks such as cell-type classification and gene expression imputation. Notes:

HV-002 | FORMAL VALIDATION **PertDiffBench: Benchmarking Diffusion Models for Single-Cell Perturbation Response Prediction** 2026 | DOI: 10.64898/2026.06.13.732013 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: The evaluated encoders included scVI, STATE, Tahoe-x1, scGPT, scFoundation, 659 Geneformer and Scimilarity. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-035 | FORMAL VALIDATION **Revealing the Best Strategies for Rare Cell Type Detection in Multi-Sample Single-Cell Datasets** 2025 | Genes DOI: 10.3390/genes17010031 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: while SCISSORS, scGPT+IF showed minimal changes across all conditions (−0.04 to +0.04;−0.05 to +0.04). Notes:
HV-011 | FORMAL VALIDATION **NanoCellAnnotator: Formalizing Expert Cell Type Annotation with Large Language Models** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.21.728965 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Furthermore, many LLM-based annotation pipelines depend on large cloud-hosted models, limiting reproducibility and practical deployment in clinical or privacy-sensitive settings [4] [5]. Notes:
HV-045 | FORMAL VALIDATION **Opportunities for artificial intelligence and synthetic biology in designing living drug delivery systems** 2026 | Advanced Drug Delivery Reviews DOI: 10.1016/j.addr.2026.115883 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Absent Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-020 | FORMAL VALIDATION **Challenges in AI-driven Biomedical Multimodal Data Fusion and Analysis** 2025 | Genomics Proteomics & Bioinformatics DOI: 10.1093/gpbjnl/qzaf011 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: scGPT [14] focuses on processing these high-dimensional and sparse RNA data to capture complex relationships between cells, thereby supporting transfer learning across datasets and even across species. Notes:
HV-023 | FORMAL VALIDATION **Genetic Convergence Analysis of CRISPR Perturbations Deciphers Gene Functional Similarity** 2025 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.11.13.688060 Article page Full text 1 Full text 2	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Generative models have been developed to predict the outcome of perturbation experiments that have not been performed [52, 53, 54]. Notes:

HV-039 | FORMAL VALIDATION **scGNN+: Adapting ChatGPT for Seamless Tutorial and Code Optimization** 2024 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.09.30.615735 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Foundation models1 have revolutionized AI by using large-scale data to perform diverse tasks with exceptional efficiency. In bioinformatics, these models have been applied to single-cell RNA sequencing (scRNA-seq) and high-throughput sequencing data, primarily focusing on datacentric advantages such as enhanced comprehensive analysis in accuracy and scalability2-5. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-034 | FORMAL VALIDATION **GET: a foundation model of transcription across human cell types** 2023 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2023.09.24.559168 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: For example, recent developments in single cell foundation models like Geneformer11, scGPT12, and scFoundation13 have demonstrated how diverse transcriptome profiles can be encoded in one model, enabling various contextualized downstream tasks including cell type annotation and perturbation prediction Notes:
HV-031 | FORMAL VALIDATION **C5aR1 inhibition reprograms tumor associated macrophages and reverses PARP inhibitor resistance in breast cancer** 2024 | Nature Communications DOI: 10.1038/s41467-024-48637-y Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Apply to biological data Biological insight: Yes Evidence sufficiency: Sufficient Supporting quote + locator: For further clari cation of macrophage pheno- types, a reverse annotation was done by scGPT with MDST datasets as a training model and a large human myeloid dataset as a test data Notes:

HV-040 | FORMAL VALIDATION **Clifti-GPT: Privacy-preserving federated fine-tuning and transferable inference of foundation models on clinical single-cell data** 2025 | DOI: 10.21203/rs.3.rs-7917089/v1 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Built upon the scGPT foundation model, clifti-GPT achieves performance within 4% of centralized baselines in accuracy, precision, recall, and macro-F1 for cell type classification and reference mapping across six datasets. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-001 | FORMAL VALIDATION **Generative genomics accurately predicts future experimental results** 2025 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.09.08.674753 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We compared to embeddings from scGPT19, a widely used, pretrained model of scRNA-seq, for a reference point. GEM-1's biological embedding yielded a stronger separation of cells by cell type than did GEM-1's technical embedding, as expected, and was qualitatively similar to scGPT's embedding For scGPT19 (Fig. 4A, bottom row), embeddings were extracted by passing observed single-cell expression profiles through the encoder of the pretrained scgpt_full model, again producing cell-level representations. Embeddings were obtained using the scGPT library tasks.embed_data function from scGPT version 0.2.4. We then visualized the data similarly to the bulk RNA-seq embeddings. Notes:
HV-038 | FORMAL VALIDATION **cspray: Distributed Single Cell Transcriptome Analysis** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.06.704110 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: On the first point, one may consider foundation models for single cells (see e.g. [9–12]) to be a perfect fit for cell type labeling across many samples. Indeed, running large scale inference of foundation models is inherently scalable. These models still typically require one to prepare the input data, which can involve out of memory (OOM) issues for large datasets. Notes:

HV-021 | FORMAL VALIDATION **scaLR: a low-resource deep neural network-based platform for single cell analysis and biomarker discovery** 2025 | Briefings in Bioinformatics DOI: 10.1093/bib/bbaf243 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Recently, deep neural network (DNN) based models have been introduced to enhance the accuracy and scalability of cell type annotation [5–7]. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-047 | FORMAL VALIDATION **A technical review of multi-omics data integration methods: from classical statistical to deep generative approaches** 2025 | Briefings in Bioinformatics DOI: 10.1093/bib/bbaf355 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Similarly, scGPT [180], a generative pre-trained transformer trained on over 33 million single-cell transcriptomic profiles, supports batch correction, perturbation modelling, gene network inference, and is designed to extend toward multi-omics integration. Notes:
HV-015 | FORMAL VALIDATION **scGPT-spatial: Continual Pretraining of Single-Cell Foundation Model for Spatial Transcriptomics** 2025 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.02.05.636714 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We introduce scGPT-spatial, a specialized foundation model for spatial transcriptomics continually pretrained on our previously published scGPT scRNA-seq foundation model. Notes:

HV-017 | FORMAL VALIDATION **CellMemory: hierarchical interpretation of out-of-distribution cells using bottlenecked transformer** 2025 | Genome biology DOI: 10.1186/s13059-025-03638-y [Article page]() [Full text 1]()	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: In comparison with advanced integration tools [37] such as scVI [38], scGPT [16], and scPoli [6], which is tailored for the integration of population-scale datasets. CellMemory consistently preserved biological represen-tations while minimizing batch effects (Fig. 2c, Additional file 1: Fig. S5). Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-029 | FORMAL VALIDATION **Elucidating the Design Space of Generative Models for Single-Cell Perturbation Prediction** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.15.732063 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: *Comparison with scGPT* On the same benchmark, scGPT (6) achieves Enrichment AUC = 0.79. Our strongest Parse-trained configuration reaches 0.79, within typical evaluation noise of scGPT despite scGPT consuming roughly 10× more pre-training data. The point is not a SOTA claim on the TeloHAEC benchmark, but a representation-efficiency claim: a discrete-latent perturbation prior trained on a single 1M-cell dataset effectively matches a foundation model trained on tens of millions of cells, on a biologically meaningful ranking task. The explicit zero-gate decoder identified by our ablation transfers cleanly off the training distribution. We acknowledge that this benchmark has several distribution shifts such as cell type, perturbation set, biological context, etc, making it hard to identify which axis of generalization the model actually uses with the limited training dataset. Notes:

HV-004 | FORMAL VALIDATION **Discovering Candidate Anti-Aging Perturbations Using a Foundation Model for Gene Expression** 2025 | International Journal of Molecular Sciences DOI: 10.3390/ijms262411977 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Apply to biological data Biological insight: Yes Evidence sufficiency: Sufficient Supporting quote + locator: Using human single-cell RNA-seq data from the multi-tissue AgeAnno dataset, we fine-tuned scGPT, a large transcriptomic model, to predict chronological age groups, achieving high classification accuracy. To identify genes influencing age predictions, we systematically perturbed individual genes in silico and quantified their effects, classifying them as pro- or anti-aging candidates. Our results demonstrate that scGPT does capture age-related dependencies in single-cell data and can be utilized to discover novel candidate gene perturbations—potential targets to be validated as anti-aging interventions. Notes:
HV-013 | FORMAL VALIDATION **Integrating In silico perturbation with multilayer omics to decode regulatory networks in cancer immunity: a new frontier in precision oncology** 2026 | Frontiers in Immunology DOI: 10.3389/fimmu.2026.1857447 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: While early deep learning frameworks provided a starting point, current trends involve models like scGPT and GearNet, which utilize pre-trained cell embeddings to learn the language of gene expression. A defining advantage of these foundation models is their dual capability for zero-shot and fine-tuned perturbation predictions. In a zero-shot setting, models such as scGPT leverage their vast pre-training on diverse cell atlases to predict the effects of unseen perturbations exemplified by the knockout of genes not explicitly included in the training data by navigating the learned latent space of gene regulation (30). Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-007 | FORMAL VALIDATION **Benchmarking gene expression reconstruction from single-cell latent representations** 2026 | DOI: 10.64898/2026.06.15.731445 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: An end-to-end reconstruction, we compare PCA, AEs, and VAEs, which are widely used in single-cell genomics and machine learning more broadly. In foundation-model reconstruction, we evaluate embeddings learned from four foundation models (SE embedding from STATE1, scConcept, scGPT, SCimilarity)17,24–26 paired with three decoder architectures (multilayer perceptron (MLP), k-nearest neighbors (KNN), and Transformer). Notes:
HV-025 | FORMAL VALIDATION **Hybrid Attention-Enhanced Regularized Convolutional Autoencoder for Robust EEG Feature Extraction in Seizure Prediction** 2025 | IEEE Access DOI: 10.1109/access.2025.3608302 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: In addition to seizure prediction, the quality of the extracted features was assessed utilizing unsupervised clustering with the Leiden algorithm [44]. Moreover, feature quality was evaluated based on biological conservation and technical batch correction, consistent with several clustering performance metrics adopted in a previous study [45]. Notes:

HV-041 | FORMAL VALIDATION **Nicheformer: a foundation model for single-cell and spatial omics** 2025 | Nature Methods DOI: 10.1038/s41592-025-02814-z Article page Full text 1 Full text 2	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Comparisons against Geneformer, scGPT, UCE and CellPLM To get the Geneformer embeddings, we used the release v.0.0.1 of the official Geneformer repository on Hugging Face and extracted the embeddings using the pretrained weights of the larger 12-layer variant provided at the time. We used the second to last layers to get a more general representation as recommended by the repository. We also used mean pooling as the only available option provided to aggregate the output gene embeddings into a single-cell embedding. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-033 | FORMAL VALIDATION **scCCVGBen for benchmarking of single-cell representation learning anchored on a centroid-coupled variational graph attention autoencoder across scRNA-seq and scATAC-seq** 2026 | Frontiers in Genetics DOI: 10.3389/fgene.2026.1822168 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: scGPT (Cui et al., 2024) and scFoundation (Hao et al., 2024) are pre-trained on tens of millions of cells and exposed via inference paths that are atlas-scoped rather than per-dataset. Rather than re-fitting them per cohort dataset, which conflicts with the pre-training contract, we evaluated their published checkpoints in the same controlled, per-dataset paired protocol used for every other comparator, as shown in Figure 7. Cell-level latents were obtained via scgpt.tasks.embed_data on the whole-human checkpoint with use_fast_transformer=False and via scfoundation.model.get_embedding with cell-level output and tgthighres=f1. The mouse cohort (GSE226131) initially returned only 17 of 31,053 gene-symbol matches against the scGPT human vocabulary; we exposed orthologs by upper-casing the mouse gene symbols at inference time, lifting the match to 15,935 of 31,053 genes. Notes:
HV-008 | FORMAL VALIDATION **Towards Cross-Sample Alignment for Multi-Modal Representation Learning in Spatial Transcriptomics** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.03.02.709002 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: To address some of these limitations, recent transcriptomics foundation models have been developed to generate richer non-linear embeddings that capture subtle transcriptomics patterns Cui et al. (2024a;b); Hao et al. (2024). B Notes:

HV-009 | FORMAL VALIDATION **iPro-CSAF: identification of promoters based on convolutional spiking neural networks and spiking attention mechanism** 2025 | PeerJ Computer Science DOI: 10.7717/peerj-cs.2761 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: In bioinformatics fields researchers have developed many machine learning methods and especially deep learning methods to help solving problems in many fields including gene expression (Al taweraqi & King, 2022), protein structures (AlQuraishi, 2021; Tubiana, Schneidman-Duhovny & Wolfson, 2022) genomic (Dalla-Torre et al., 2024) and single-cell biology (Cui et al., 2024), etc. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-036 | FORMAL VALIDATION **CART-GPT: A T Cell-Informed AI Linguistic Framework for Interpreting Neurotoxicity and Therapeutic Outcomes in CAR-T Therapy** 2025 | DOI: 10.1101/2025.08.08.669387 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop Biological insight: Yes Evidence sufficiency: Sufficient Supporting quote + locator: architecture of CART-GPT. The model builds upon the pretrained scGPT by freezing its embedding layers and fine-tuning the remaining components to create TcellGPT, a general-purpose model for T cell annotation. TcellGPT was then further trained on CAR-T data to generate CART-GPT-response for predicting therapy response. Thus, TcellGPT was subsequently fine-tuned into CART-GPT-ICANS. Notes:
HV-050 | FORMAL VALIDATION **Context-dependent utility and robustness of pretrained single-cell foundation model representations across analytical tasks** 2026 | DOI: 10.64898/2026.06.18.733285 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: scGPT ranked within the top five in five of the six evaluated metrics and achieved the highest score in kBET, demonstrating broad competitiveness across integration criteria. Notes:

HV-006 | FORMAL VALIDATION **Artificial Intelligence agents for biological research: a survey** 2026 | Briefings in Bioinformatics DOI: 10.1093/bib/bbag075 [Article page] [Full text 1]	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: This article aims to survey the progress on recent agent-based studies in biological research. To better understand their conceptual and methodological uniqueness, it is necessary to contrast them with foundation models which currently dominate biological modeling. Foundation models such as scGPT [19], scFoundation [22], Geneformer [23], and GeneMamba [24] establish a powerful and robust framework for modeling high-dimensional biological data. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-032 | FORMAL VALIDATION **OCOO-T : A SIMPLE AND SCALABLE VIRTUAL CELL MODEL FOR TRANSCRIPTIONAL PERTURBATION RESPONSE PREDICTION** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.08.731000 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Earlier efforts to model transcriptional perturbation responses drew on the success of pre-trained language models. scGPT [1] pre-trains a generative Transformer on over 33 million human cells with a novel autoregressive gene-masking scheme, demonstrating strong transfer learning across diverse single-cell tasks. However, because single-cell RNA sequencing is destructive, true cell-level correspondences before and after perturbation are fundamentally unobservable. Perturbation response prediction is therefore an inherently distributional problem, requiring models to capture population-level shifts rather than deterministic cell-to-cell mappings. Point-wise prediction objectives, as employed by scGPT and related foundation models, are ill-suited to this setting. Notes:
HV-014 | FORMAL VALIDATION **Residual-stream geometry of single-cell foundation models carries incremental gene-regulatory signal across tissues** 2026 | BMC Bioinformatics DOI: 10.1186/s12859-026-06538-5 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: We systematically investigated residual-stream geometry in scGPT and Geneformer across four tissue contexts from the Tabula Sapiens atlas, evaluating whether geometric proximity between gene vectors provides incremental predictive value for curated TRRUST transcription factor–target edges beyond expression confounds. Notes:

HV-037 | FORMAL VALIDATION **Tahoe-x1: Scaling Perturbation-Trained Single-Cell Foundation Models to 3 Billion Parameters** 2025 | DOI: 10.1101/2025.10.23.683759 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Here we present Tahoe-x1 (Tx1), a 3-billion-parameter single-cell foundation model family derived from a scGPT-style encoder architecture [2] and specialized for cancer therapeutics research. Tx1 is trained to jointly model the relationships between genes, pathways, cellular contexts, and chemical perturbations, with particular emphasis on predicting gene essentiality and on producing gene representations that are informative of hallmark pathway membership. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-016 | FORMAL VALIDATION **Investigation of cell development and tissue structure network based on natural Language processing of scRNA-seq data** 2025 | Journal of Translational Medicine DOI: 10.1186/s12967-025-06263-2 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: While more advanced contextual embedding methods based on Transformer architectures, such as scBERT [31], Geneformer [32], and scGPT [33], have emerged in recent years, they typically require higher computational resources and more complex model structures. Notes:
HV-024 | FORMAL VALIDATION **Scouter predicts transcriptional responses to genetic perturbations with large language model embeddings** 2025 | Nature Computational Science DOI: 10.1038/s43588-025-00912-8 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: To benchmark Scouter against gene expression foundation models—scGPT, scELMo and scFoundation—we followed the official pipelines provided by their authors (refs. 21, 22 and 23, respectively) and evaluated all models on the Adamson and Norman datasets. Notes:

HV-049 | FORMAL VALIDATION **Routine FFPE sections support clinically compatible single-nucleus transcriptomics across six human cancer types** 2026 | DOI: 10.64898/2026.07.23.740343 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Apply to biological data / Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: To enable prospective sample-by-sample analysis, data were processed using OneMap, that leverages scGPT23 foundation-model embeddings to classify cellular populations from individual samples. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-042 | FORMAL VALIDATION **Foundation model reveals the shared organization of transcription and topologically associating domains** 2026 | Cell Systems DOI: 10.1016/j.cels.2026.101675 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Extend or develop Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Contextual transcriptional similarity (CTS) applies single-cell foundation models (scFMs) to a new purpose: detecting gene relationships that co-expression misses. Derived from scGPT (single-cell Generative Pre-trained Transformer), a foundation model trained on 33 million cells, CTS captures whether genes occupy similar roles within cellular programs. Notes:
HV-003 | FORMAL VALIDATION **sciLaMA: A Single-Cell Representation Learning Framework to Leverage Prior Knowledge from Large Language Models** 2025 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2025.01.28.635153 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Scatter plot directly comparing sciLaMAGPT (y-axis) to sciLaMA (s.i.) (x-axis, b) and fine-tuned scGPT (x-axis, c). Notes:

HV-026 | FORMAL VALIDATION **Cancer-associated fibroblasts enhance colorectal cancer lymphatic metastasis via CLEC11A/LGR5-mediated WNT pathway activation** 2025 | Journal of Clinical Investigation DOI: 10.1172/jci194243 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Apply to biological data Biological insight: Yes Evidence sufficiency: Sufficient Supporting quote + locator: Inspired by the capabilities of foundation models in single-cell representation, particularly their exceptional performance in cross-task transfer and few-shot learning, here we applied a pretrained single-cell model, scGPT (35), to generate cell embeddings. These LLM-derived embeddings serve as reliable feature inputs for constructing machine-learning classifiers, particularly for low-confidence cells. Notes: Hard

Article and approved evidence links	Reviewer 2 independent record
HV-019 | FORMAL VALIDATION **Toward automated and explainable high-throughput perturbation analysis in single cells** 2025 | Patterns DOI: 10.1016/j.patter.2025.101228 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Additionally, the rise of virtual cell models powered by generative AI is revolutionizing medical research by enabling in silico simulations of cellular behavior. These models can predict the effects of chemical and genetic mutations, offering computational insights into biological processes.7 Notes:
HV-046 | FORMAL VALIDATION **AUPRC: a metric for evaluating the performance of in-silico perturbation methods in identifying differentially expressed genes** 2025 | Briefings in Bioinformatics DOI: 10.1093/bib/bbaf426 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: . Most recently, works that incorporate transformer based large language models like scGPT [22] leverage generative pre-training to learn biological embeddings for predicting genetic perturbation responses, annotating cell types, and integrating multi-omic datasets. Notes:

HV-027 | FORMAL VALIDATION **Bioinformatics Copilot 2.0 for Transcriptomic Data Analysis** 2024 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.08.15.607673 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Additionally, foundational models such as scFoundation7 and scGPT8, pretrained on extensive single-cell transcriptomic datasets, are being used to facilitate cell type annotation, perturbation prediction, and multi-omic integration. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-043 | FORMAL VALIDATION **DECIPHER for learning disentangled cellular embeddings in large-scale heterogeneous spatial omics data** 2025 | Nature Communications DOI: 10.1038/s41467-025-63140-8 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Embedding, or the procedure of representing original high-dimensional spatial data as low-dimensional vectors, has been widely adopted to enrich biologically relevant signals while simultaneously removing technical artifacts6,7. Notes:
HV-018 | FORMAL VALIDATION **Cellfm-datasets: A Unified Data Infrastructure for Single-Cell and Spatial Transcriptomics Foundation Model Pretraining** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.06.11.731508 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Cell foundation models transfer the pretraining paradigm from language and vision to tran-scriptomic profiles. Representative systems include scBERT [8], Geneformer [9], scGPT [10], scFoundation [11], CellFM [12], and Nicheformer [13]. These models differ architecturally, but all depend on robust out-of-core sampling over sparse, heterogeneous, and repeatedly reused training corpora. Notes:

HV-030 | FORMAL VALIDATION **A Foundation Model Should Not Flatten Pharmaceutical Evidence across Molecules, Sequences, Omics, Images, and Scientific Text** 2024 | Pharmacophore DOI: 10.51847/rbfe0o1wzp Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: No Primary use: Mention only Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: Protein-structure prediction depends on architectures and priors suited to sequence, evolutionary, and geometric information; transcriptomic transfer learning depends on cell-state and network contexts; and single-cell foundation modeling uses gene- and cell-specific objectives to support multi-omic, perturbational, and annotation tasks [17-19]. Notes:

Article and approved evidence links	Reviewer 2 independent record
HV-010 | FORMAL VALIDATION **evoCancerGPT: Generating Zero-Shot Single-Cell and Single-Sample Cancer Progression Through Transfer Learning** 2026 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.64898/2026.02.12.705621 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: evoCancerGPT captures key transitions in cancer evolution, achieving high concordance to ground truth trajectories and outperforming linear and scGPT baselines in held-out test samples in low-context scenarios. Notes:
HV-044 | FORMAL VALIDATION CellAgent: LLM-Driven Multi-Agent Framework for Natural Language-Based Single-Cell Analysis 2024 | bioRxiv (Cold Spring Harbor Laboratory) DOI: 10.1101/2024.05.13.593861 Article page Full text 1	Review status: LOCKED Citation validity: Yes Method executed: Yes Primary use: Benchmark Biological insight: No Evidence sufficiency: Sufficient Supporting quote + locator: To evaluate the effectiveness of CellAgent in cell type annotation, we compared the results of CellAgent and other methods, such as scGPT [10], CellMarker [43], and Cell-Typist [44], among others, against the manual annotations provided by the original studies across six datasets. Notes: